Generalized Inverses, Projections, and Least-Squares Geometry

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§1.1 What an inverse was supposed to do

For a square nonsingular matrix, the inverse performs two jobs at once. It solves every equation

$$Ax = b$$

and it undoes the action of A:

$$A^{-1}A = I, \qquad AA^{-1} = I.$$

Once A is rectangular or rank deficient, these two jobs separate. Some vectors b may not lie in the column space, so an exact solution does not exist. If the null space is nontrivial, several vectors x may produce the same output, so an exact solution need not be unique.

The Moore–Penrose inverse does not pretend that these failures have disappeared. It makes two geometric choices:

(i) replace an unreachable target by its orthogonal projection onto the column space;

(ii) among all inputs producing that projected target, choose the one orthogonal to the null space.

The resulting vector

$$x^+ = A^+ b$$

is therefore canonical even when ordinary inversion is impossible.

The familiar cases are recovered as special formulas:

matrix type	formula for A^+	interpretation
$m = n = \operatorname{rank} A$	A^{-1}	ordinary inverse
$m > n,\ \operatorname{rank} A = n$	$(A^*A)^{-1}A^*$	full-column-rank least squares
$m < n,\ \operatorname{rank} A = m$	$A^*(AA^*)^{-1}$	full-row-rank minimum norm
$\operatorname{rank} A < \min(m, n)$	$V\Sigma^+U^*$	rank-deficient case

The last line is the real definition in computational practice; the first three are consequences of it.

§1.2 The four fundamental subspaces reveal the answer

Let

$$A : \mathbb{C}^n \longrightarrow \mathbb{C}^m.$$

The domain and codomain split orthogonally as

$$\mathbb{C}^n = R(A^*) \oplus N(A),$$

and

$$\mathbb{C}^m = R(A) \oplus N(A^*).$$

These are the four fundamental subspaces of the matrix.

The restriction

$$A\big|_{R(A^*)} : R(A^*) \longrightarrow R(A)$$

is a bijection. It is injective because

$$R(A^*) \cap N(A) = \{0\},$$

and the two spaces have the same dimension, namely $\operatorname{rank} A$. Thus A has a genuine inverse after we discard precisely the directions on which it vanishes and restrict the target to the vectors it can actually reach.

This gives a coordinate-free description of the pseudoinverse. Decompose

$$b = b_R + b_\perp, \qquad b_R \in R(A), \quad b_\perp \in N(A^*).$$

Then $A^+ b$ is the unique vector in $R(A^*)$ satisfying

$$A(A^+ b) = b_R.$$

The orthogonal component $b_\perp$ is ignored because no input can produce it.

The two central products are now transparent:

$$AA^+ = P_{R(A)}, \qquad A^+ A = P_{R(A^*)}.$$

The first projects targets onto the reachable output space. The second projects inputs away from the null-space ambiguity.

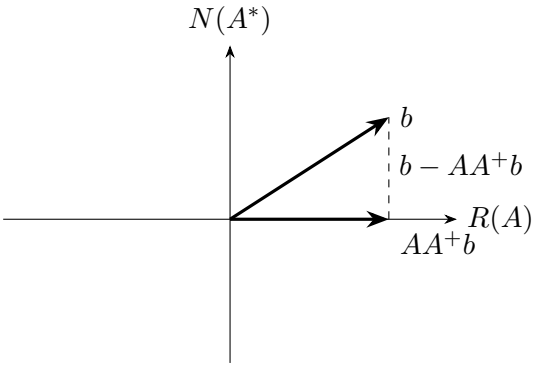


Figure 1.1: The pseudoinverse first replaces b by the closest reachable vector AA^+b.

§1.3 Why the four Penrose equations appear

The Moore–Penrose inverse is usually introduced as the unique matrix X satisfying

$$AXA = A, \qquad XAX = X,$$

$$(AX)^* = AX, \qquad (XA)^* = XA.$$

Read without context, these equations look ceremonial. The projection picture explains every one of them.

If $X = A^+$, then AX should be the orthogonal projection onto $R(A)$. An orthogonal projection is idempotent and self-adjoint. The relation

$$AXA = A$$

says that AX fixes every vector already in $R(A)$, while

$$(AX)^* = AX$$

forces the projection to be orthogonal rather than oblique.

Similarly, XA should be the orthogonal projection onto $R(A^*)$. The equations

$$XAX = X, \qquad (XA)^* = XA$$

encode that requirement on the input side.

Dropping some of the four conditions produces broader families of generalized inverses. There are 15 nonempty subsets of the four equations, which explains the traditional count of generalized-inverse classes. The count itself is less important than the distinction: algebraic retraction conditions alone allow many choices, while the two adjoint conditions select the orthogonal geometry and make the answer unique.

Proposition 1.3.1

The Moore–Penrose inverse, if it exists, is unique.

One way to see uniqueness is to use the two projection identities. Any matrix satisfying the four equations must act as the inverse of A from $R(A)$ to $R(A^*)$, must vanish on $N(A^*)$, and must return values orthogonal to $N(A)$. Those requirements determine its value on the orthogonal decomposition

$$\mathbb{C}^m = R(A) \oplus N(A^*).$$

§1.4 SVD turns the definition into a calculation

Suppose

$$A = U\Sigma V^*$$

is a singular value decomposition, with positive singular values

$$\sigma_1 \geq \cdots \geq \sigma_r > 0.$$

Write

$$\Sigma = \begin{bmatrix} \operatorname{diag}(\sigma_1, \ldots, \sigma_r) & 0 \\ 0 & 0 \end{bmatrix}.$$

Then

$$A^+ = V\Sigma^+ U^*,$$

where

$$\Sigma^+ = \begin{bmatrix} \operatorname{diag}(\sigma_1^{-1}, \ldots, \sigma_r^{-1}) & 0 \\ 0 & 0 \end{bmatrix}^T.$$

Only the nonzero singular values are inverted.

The formula is easier to remember mode by mode. Since

$$Av_i = \sigma_i u_i,$$

the pseudoinverse must satisfy

$$A^+ u_i = \frac{1}{\sigma_i} v_i \qquad (i \le r),$$

and

$$A^+ u = 0 \qquad (u \in N(A^*)).$$

Thus A^+ literally reverses the active singular directions and does nothing to unreachable directions.

The projection formulas follow at once:

$$AA^+ = U_r U_r^*, \qquad A^+ A = V_r V_r^*.$$

Here the columns of U_r form an orthonormal basis of $R(A)$, and the columns of V_r form an orthonormal basis of $R(A^*)$.

A rank factorization also gives a useful formula. If

$$A = FG,$$

where $F \in \mathbb{C}^{m\times r}$ has full column rank and $G \in \mathbb{C}^{r\times n}$ has full row rank, then

$$A^+ = G^*(GG^*)^{-1}(F^*F)^{-1}F^*.$$

The middle two inverses are only $r \times r$. This expression makes the factor-through-the-image viewpoint explicit, although SVD is usually the cleaner numerical language when rank is uncertain.

§1.5 One rank-one matrix contains the whole geometry

Let

$$u = \begin{bmatrix} 1 \\ 2 \\ 0 \end{bmatrix}, \qquad v = \begin{bmatrix} 1 \\ 1 \end{bmatrix}, \qquad A = uv^T = \begin{bmatrix} 1 & 1 \\ 2 & 2 \\ 0 & 0 \end{bmatrix}.$$

For a nonzero rank-one matrix uv^*,

$$(uv^*)^+ = \frac{vu^*}{\|u\|_2^2 \|v\|_2^2}.$$

Since

$$\|u\|_2^2 = 5, \qquad \|v\|_2^2 = 2,$$

we obtain

$$A^+ = \frac{1}{10}\begin{bmatrix} 1 & 2 & 0 \\ 1 & 2 & 0 \end{bmatrix}.$$

The two products are

$$AA^+ = \frac{1}{5}\begin{bmatrix} 1 & 2 & 0 \\ 2 & 4 & 0 \\ 0 & 0 & 0 \end{bmatrix} = \frac{uu^T}{u^T u},$$

and

$$A^+A = \frac{1}{2}\begin{bmatrix}1 & 1\\ 1 & 1\end{bmatrix} = \frac{vv^T}{v^Tv}.$$

They are the orthogonal projections onto $\mathrm{span}(u)$ and $\mathrm{span}(v)$, respectively.

Now take

$$b = \begin{bmatrix}1\\ 0\\ 1\end{bmatrix}.$$

The closest reachable vector is

$$\widehat{b} = AA^+b = \frac{1}{5}\begin{bmatrix}1\\ 2\\ 0\end{bmatrix}.$$

The residual is

$$r = b - \widehat{b} = \begin{bmatrix}\frac{4}{5}\\ -\frac{2}{5}\\ 1\end{bmatrix},$$

and indeed

$$u^Tr = 0.$$

The pseudoinverse solution is

$$x^+ = A^+b = \begin{bmatrix}\frac{1}{10}\\ \frac{1}{10}\end{bmatrix}.$$

Every vector satisfying

$$x_1 + x_2 = \frac{1}{5}$$

produces $\widehat{b}$, but x^+ is the one with smallest Euclidean norm. The equality of its two coordinates is not accidental: it is the orthogonal projection of the origin onto that affine line.

This single example displays all four subspaces. The column space is $\mathrm{span}(u)$; the left null space is its orthogonal complement. The row space is $\mathrm{span}(v)$; the null space consists of vectors proportional to $(1,-1)^T$. The pseudoinverse moves only between the two one-dimensional active spaces.

§1.6 Exact solutions, least squares, and minimum norm

For an arbitrary b, the vector

$$x^+ = A^+b$$

minimizes

$$\|Ax - b\|_2.$$

Indeed,

$$Ax^+ = AA^+b = P_{R(A)}b,$$

which is the closest point in the column space to b.

All least-squares minimizers have the form

$$x = A^+b + (I - A^+A)z, \qquad z \in \mathbb{C}^n.$$

The second term lies in $N(A)$, so it does not change Ax. Since

$$A^+b \in R(A^*) = N(A)^\perp,$$

the two terms are orthogonal. Therefore

$$\|x\|_2^2 = \|A^+b\|_2^2 + \|(I - A^+A)z\|_2^2.$$

This proves that A^+b is the unique minimum-norm least-squares solution.

Exact solvability is detected by

$$AA^+b = b.$$

If this holds, the same formula describes every exact solution:

$$x = A^+b + (I - A^+A)z.$$

Thus one expression covers the overdetermined, underdetermined, and rank-deficient cases. What changes is only whether the projection AA^+b equals b and whether the null-space term is present.

The common identities

$$(A^+)^+ = A, \qquad (A^*)^+ = (A^+)^*, \qquad \operatorname{rank} A^+ = \operatorname{rank} A$$

are immediate from the SVD. They are consequences of the singular-direction picture, not a separate collection of tricks.

§1.7 Why inverse-like algebra can fail

Ordinary inverses reverse products:

$$(AB)^{-1} = B^{-1}A^{-1}.$$

The corresponding pseudoinverse identity is false in general. Let

$$A = \begin{bmatrix} 1 & 1 \end{bmatrix}, \qquad B = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}.$$

Then

$$AB = \begin{bmatrix} 1 & 0 \end{bmatrix}, \qquad (AB)^+ = \begin{bmatrix} 1 \\ 0 \end{bmatrix}.$$

On the other hand,

$$A^+ = \frac{1}{2}\begin{bmatrix} 1 \\ 1 \end{bmatrix}, \qquad B^+ = B,$$

so

$$B^+A^+ = \begin{bmatrix} \frac{1}{2} \\ 0 \end{bmatrix} \neq (AB)^+.$$

The failure is geometric. Each pseudoinverse inserts orthogonal projections onto row and column spaces. In a product, the active output space of B need not align with the active input space of A. Reversing the two pseudoinverses introduces projections in the wrong places. Reverse-order laws do hold under additional range and commutation hypotheses, but they should never be assumed from notation alone.

This is a useful general warning: the pseudoinverse behaves like an inverse only on the active singular subspaces. Algebraic identities that move through kernels or orthogonal complements require extra compatibility.

§1.8 Updating a pseudoinverse one column at a time

Recomputing an SVD after every new data column may be wasteful. Greville's recursion updates the pseudoinverse when

$$A_{k+1} = \begin{bmatrix} A_k & a \end{bmatrix}.$$

Set

$$d = A_k^+ a, \qquad c = a - A_k d = (I - A_k A_k^+)a.$$

The vector c is the part of the new column orthogonal to the old column space. If $c \neq 0$, define

$$b^* = \frac{c^*}{c^* c}.$$

Then

$$A_{k+1}^+ = \begin{bmatrix} A_k^+ - d b^* \\ b^* \end{bmatrix}.$$

For a small example, begin with

$$A_1 = \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \qquad A_1^+ = \begin{bmatrix} 1 & 0 \end{bmatrix},$$

and append

$$a = \begin{bmatrix} 1 \\ 1 \end{bmatrix}.$$

Then

$$d = A_1^+ a = 1, \qquad c = a - A_1 d = \begin{bmatrix} 0 \\ 1 \end{bmatrix}, \qquad b^* = \begin{bmatrix} 0 & 1 \end{bmatrix}.$$

Therefore

$$A_2^+ = \begin{bmatrix} 1 & -1 \\ 0 & 1 \end{bmatrix}.$$

But

$$A_2 = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$$

is invertible, and the update has recovered its ordinary inverse exactly.

If $c = 0$, the new column already lies in the old column space and a different scalar correction is used. That branch is not a technical nuisance; it records whether the incoming data adds a genuinely new direction.

§1.9 Iteration sees the pseudoinverse as a limit

The pseudoinverse need not be formed explicitly in a large problem. Consider Landweber iteration

$$x_{k+1} = x_k + \alpha A^*(b - A x_k),$$

with

$$0 < \alpha < \frac{2}{\sigma_1^2}.$$

This is gradient descent on

$$\frac{1}{2}\|Ax - b\|_2^2.$$

In singular-vector coordinates, each active component evolves independently. If

$$x_k = \sum_i \xi_{i,k} v_i,$$

then

$$\xi_{i,k+1} = (1 - \alpha\sigma_i^2)\xi_{i,k} + \alpha\sigma_i u_i^* b.$$

The step-size condition makes

$$|1 - \alpha\sigma_i^2| < 1$$

for every nonzero singular value. Hence

$$\xi_{i,k} \longrightarrow \frac{u_i^* b}{\sigma_i}.$$

If the iteration starts from $x_0 = 0$, the null-space components remain zero, so

$$x_k \longrightarrow A^+ b.$$

The minimum-norm choice is therefore encoded by the initialization: gradient descent never invents a component in a direction that the data cannot see.

This observation is useful beyond numerical linear algebra. In an underdetermined linear model, the optimization dynamics themselves can impose an implicit bias toward the minimum-norm solution.

§1.10 A tiny noisy direction: three spectral filters

Take a matrix which almost erases its second coordinate:

$$A = \begin{pmatrix} 1 & 0 \\ 0 & 10^{-2} \end{pmatrix}.$$

Suppose the exact signal is $x_* = (1, 0)^T$, so $Ax_* = (1, 0)^T$, but we observe

$$b = \begin{pmatrix} 1 \\ 0.02 \end{pmatrix}.$$

The perturbation of the data is only 0.02 in norm. Exact pseudoinversion turns it into

$$\boxed{A^+ b = (1, 2)^T,}$$

an error of size 2 in the recovered vector. The small singular direction has amplified the noise a hundredfold.

A Tikhonov penalty with $\lambda = 10^{-3}$ changes the answer to

$$x_\lambda = \begin{pmatrix} \dfrac{1}{1 + 10^{-3}} \\ \dfrac{10^{-2}}{10^{-4} + 10^{-3}}\, 0.02 \end{pmatrix} = \begin{pmatrix} \dfrac{1000}{1001} \\ \dfrac{2}{11} \end{pmatrix}.$$

It suppresses the nearly invisible coordinate, but at a cost: if the true signal *had* a large component in that direction, regularization would erase part of it. There is no universally correct cutoff without an assumption about signal and noise.

The preceding Landweber iteration hides a related kind of regularization. Starting at $x_0 = 0$, solving its scalar recurrence in each singular direction gives an exact finite-step formula:

$$\boxed{x_k = \sum_{\sigma_i > 0} \frac{1 - (1 - \alpha\sigma_i^2)^k}{\sigma_i} (u_i^* b) v_i.}$$

For $\alpha = 1$ and $k = 1000$ in our example, the first coordinate is 1, while the second is

$$2\left(1 - \left(1 - 10^{-4}\right)^{1000}\right) \approx 0.1903.$$

The iterative solver has not yet amplified that direction all the way to 2. Near $\sigma = 0$, the finite-step filter behaves as

$$\frac{1 - (1 - \alpha\sigma^2)^k}{\sigma} \sim k\alpha\sigma;$$

Tikhonov's filter behaves as σ/λ. They are different functions with the same useful feature: both attenuate the tiniest singular directions.

In this example, running the iteration *to convergence* would recover the noisy pseudoinverse $(1, 2)^T$. Stopping early is therefore a meaningful part of the inverse problem, not simply an unfinished numerical computation.